Rational Atoms after Residue Splitting: Determinant Inflation and the DFT Ledger

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July 2026

Abstract

A rational twist of conductor R splits into R residue-class sums. On each class the two affine slopes have gcd exactly R and the determinant becomes 2R. One Fourier mode is one DFT coordinate, not control of every residue sum. The classification is `ALGEBRAIC_REDUCTION_L1` and the verdict is `PROVED_RESIDUE_SPLIT_WITH_DETERMINANT_2R`. No program-positive L2 claim or endpoint exponent credit is made.

1 Target contract and source boundary

The target has six simultaneous axes: actual fixed-h_0 packet, source-locked named physical atom, every deterministic prefix, every deterministic scale, a fixed-X power at that atom, and actual active support. The value $h_0 = 2$ is source-backed data only. Repository hashes are integrity locks, not theorem evidence. TPC-193's declared seven-source corpus remains STOP-SCOPED [2].

2 Exact residue decomposition

Let $\alpha = r/R$ in lowest terms and let

$$C_r(Z) = \sum_{z \in Z} c(z) \mathrm{e}(-rz/R).$$

Partitioning by $z = b + Rm$ gives the identity

$$C_r(Z) = \sum_{b \bmod R} \mathrm{e}(-rb/R) \sum_{\substack{m \\ b+Rm \in Z}} c(b + Rm). \tag{1}$$

For the determinant-two forms $d + sz, u + az$, the two forms on the b-th class are

$$(d + sb) + sRm, \qquad (u + ab) + aRm.$$

Theorem 1 (Determinant and slope-gcd ledger)**.** *If $su - ad = 2$ and as is odd, then $(a, s) = 1$. After residue splitting the two slopes sR, aR have gcd exactly R, while their determinant is*

$$(sR)(u + ab) - (aR)(d + sb) = 2R.$$

Equation (1) is one DFT coordinate. Recovering every residue sum requires all R Fourier coordinates and DFT inversion.

Proof. Any common divisor of a, s divides 2; oddness forces it to be one. The displayed determinant follows by expansion. The final assertion is the invertibility of the $R \times R$ Fourier matrix; a single row has a kernel of dimension $R - 1$ when $R > 1$. □

3 Imported identities and new crosswalk

The DFT identity and the raw progression determinant law are imported from the TPC-108 alias calculus and TPC-94 phase ledger [3, 1]. The L1 contribution here is their joint typed crosswalk to the TPC-194 decorated direct-prefix object, including the warning that a single mode is not all residue classes.

4 Firewall

The slope-gcd-R, determinant-$2R$ residue problem is not the original primitive-slope determinant-two problem. A theorem for one rational Fourier mode also cannot be relabelled as simultaneous residue-class cancellation.

5 Loss, level and scope ledger

The smallest missing literal theorem is

`UNIFORM_ALL_RESIDUE_CLASS_CANCELLATION_OR_DIRECT_MODE_THEOREM`.

The next route is `RATIONAL_ATOM_THEOREM_SCREEN`. No new method cell is stopped in this paper.

The named-atom exponent credit is 0, while the endpoint requires a strict budget greater than 1/400; the ledger is unpaid. Block sums are not cumulative prefixes, symbolic packet atoms are not named production atoms, phase L^2 and almost-everywhere statements are not pointwise evaluation, and scoped method failures are not theorem or architecture failures.

6 Machine certificate

The adjacent canonical payload freezes the formula or finite witness, exact repository-source hashes, any explicitly non-hashed external locator record, six target axes, scoped stop, route state and claim firewall. Its recursive exact schema has closed objects, exact array positions and constant leaves. The checker recomputes the finite certificate and executes ten adversarial mutations.

References

[1] Liang Wang. Literal generic affine mobius dispersion, 2026.

[2] Liang Wang. Literal fixed-atom candidate mechanism gate, 2026.

[3] Liang Wang. Exact content resonance ledger, 2026.